

Titanic Exploratory Data Analysis

Data Analyst Internship — Task 5

1. Objective

The objective of this task is to explore the Titanic dataset using Python and identify important patterns, relationships, trends, distributions, and missing values through statistical and visual analysis.

2. Tools Used

Python, Pandas, Matplotlib, Seaborn, and Jupyter Notebook.

3. Dataset Overview

The dataset contains 891 passenger records and 12 columns. The variables include passenger ID, survival status, passenger class, name, sex, age, family information, ticket, fare, cabin, and embarkation port.

Metric	Value
Rows	891
Columns	12
Overall survival rate	25.70%
Missing Age values	177
Missing Cabin values	702
Missing Embarked values	2

4. Exploratory Analysis Performed

- Dataset structure and data types were inspected using `df.info()`.
- Statistical measures were examined using `df.describe()`.
- Missing values and duplicate records were checked.
- Survival counts were examined using `value_counts()`.
- Univariate analysis was performed using histograms and boxplots for Age and Fare.
- Bivariate analysis included gender vs survival, passenger class vs survival, Age vs Fare, and Fare vs survival.
- A correlation heatmap was used to examine relationships among numerical variables.
- A pairplot was used for multivariate visual exploration of numerical variables by survival status.
- Missing Age and Embarked values were handled using median and mode respectively.

5. Key Findings

- Female passengers had a higher survival percentage than male passengers (50.7% vs 13.3% in this dataset).
- Survival varied across passenger classes; the calculated survival rates were approximately 48.9% for Class 1, 28.2% for Class 2, and 14.5% for Class 3.
- Fare values were widely distributed and the boxplot was used to identify unusually high fare values.
- Age showed a broad distribution, with missing observations identified before imputation.
- The correlation heatmap and pairplot provided a compact view of relationships among numerical variables.

6. Visualizations Included in the Notebook

- Survival Count
- Survival by Gender
- Survival by Passenger Class
- Age Distribution
- Fare Distribution
- Age Boxplot
- Fare Boxplot
- Age vs Fare
- Fare vs Survival
- Correlation Heatmap

7. Conclusion

The EDA demonstrates how statistical summaries and visualizations can be combined to understand a dataset. The analysis highlights survival differences by gender and passenger class, examines distributions and outliers, and identifies missing data that requires preprocessing. The notebook provides a practical exploration of the Titanic dataset using Pandas, Matplotlib, and Seaborn.